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Machine Learning for Shock-Capturing Numerical Schemes

Learning reconstruction decisions for
1D hyperbolic conservation laws

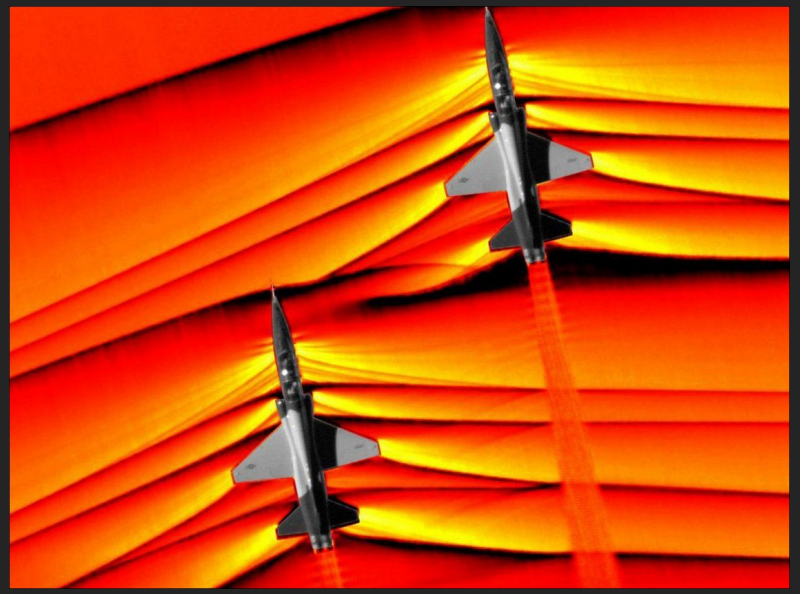
WENO-NN / NN-TENO5 / DIFFERENTIABLE CFD

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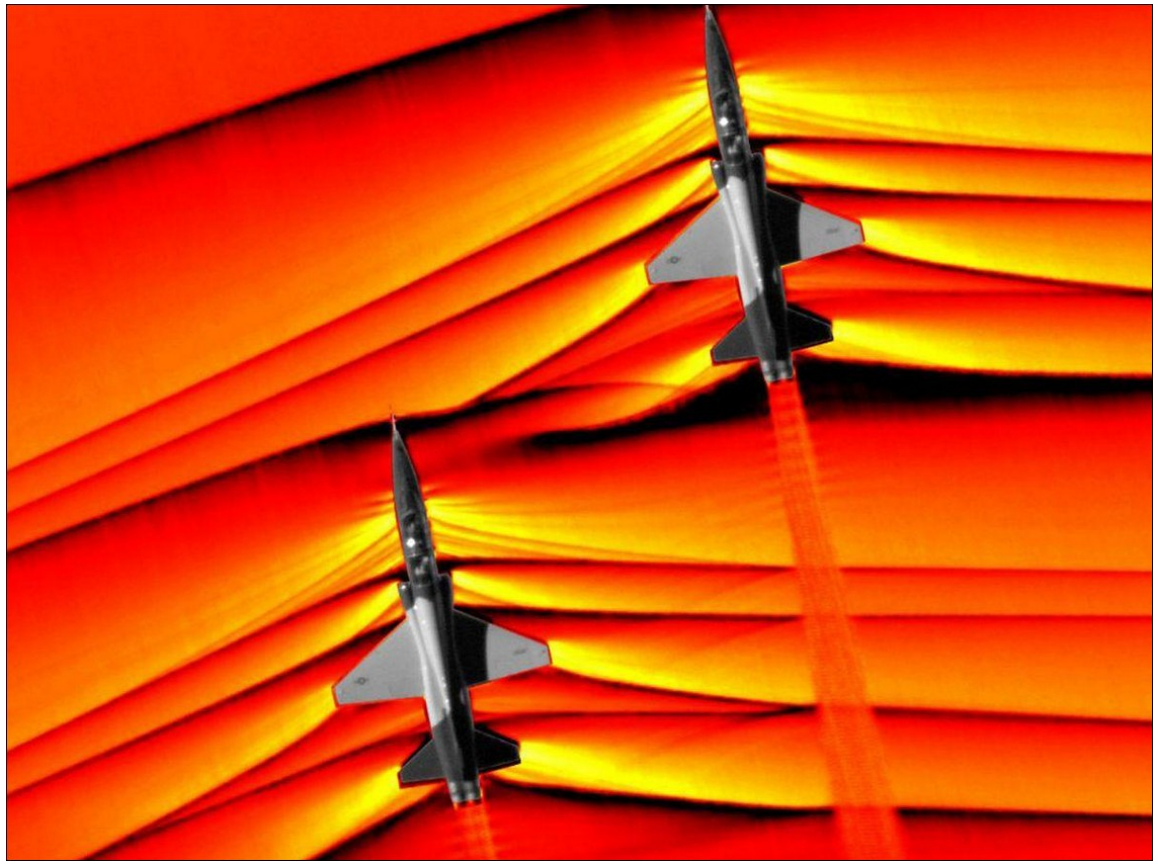
PHYSICS × NUMERICS × LEARNING



Capturing sharp fronts without spurious oscillations

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Shock capturing balances accuracy and stability



Schlieren imaging reveals abrupt changes in compressible flow.

The governing conservation law

$$\partial_t u + \partial_x f(u) = 0$$

$$f(u) = au \quad \text{or} \quad f(u) = \frac{u^2}{2}$$

Smooth regions

Resolve the solution with high-order accuracy.

Discontinuities

Suppress nonphysical oscillations near shocks.

Euler flows

Maintain positive density and pressure.

> The learned component must work within the structure of a conservative numerical solver.

The solver follows a conservative numerical path

Forward numerical path

State + local stencil → Scalar / Euler setup → **WENO / TENO or NN selector** → Interface flux → SSPRK3 step → K-step rollout

Learned component

Conservation through shared fluxes

$$\frac{d\bar{u}_i}{dt} = -\frac{\hat{f}_{i+1/2} - \hat{f}_{i-1/2}}{\Delta x}$$

Each cell changes by its incoming and outgoing flux.

Advance the solution in time

SSPRK3 uses three stages per time step.
A K-step rollout repeats the complete solver.
The time step respects the CFL condition.

> The network changes the reconstruction decision inside a conservative solver.

Euler equations describe compressible flow

The system tracks three conserved quantities throughout the flow.

Density

How mass is distributed in the flow.

Momentum

How the motion of the fluid is transported.

Total energy

Internal and kinetic energy together.

State used by the solver

$$U = (\rho,\ \rho v,\ E)^T$$

ρ: density v: velocity E: total energy

Physical states must remain valid

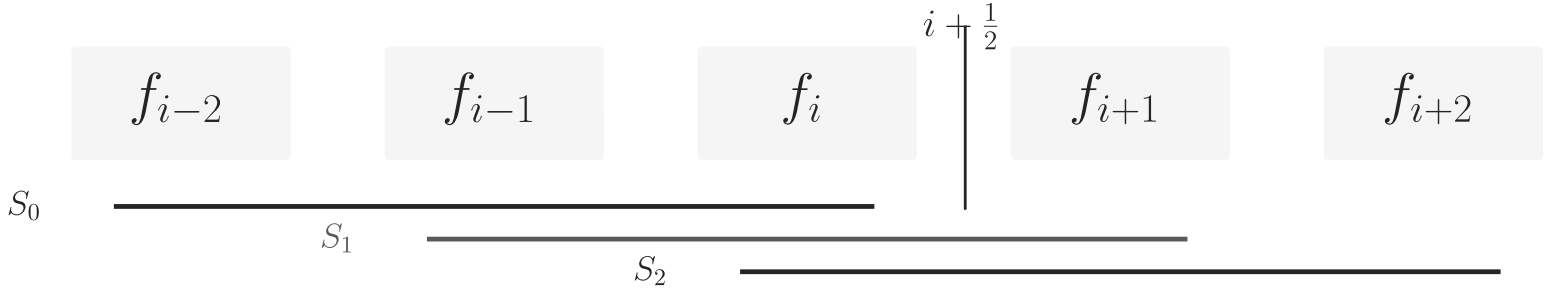
$$\rho > 0, \qquad p > 0$$

Density and pressure must remain positive.

> Sod and Shu–Osher test the learned reconstruction in compressible Euler flows.

WENO and TENO combine three local reconstructions

Five stencil values → three third-order candidates → one interface flux



ENO selects

Choose the smoothest candidate.

Keep one reliable candidate.

WENO weights

Reduce troubled candidates continuously.

Smooth region:
spread trust across the candidates.

TENO discards

Reject troubled candidates, then reweight.

Near a shock:
reject a candidate crossing the jump.

$$\hat{f}_{i+1/2} = \omega_0 q_0 + \omega_1 q_1 + \omega_2 q_2$$

Combine with local weights.

The network predicts which candidates to trust

A local five-point stencil feeds a classifier with three troubled-candidate probabilities.

Normalize the input

Scale the local values before classification.

Constant stencils receive a safe, fixed input.

→

FEED-FORWARD CLASSIFIER

5 → **64** → **32** → **16** → **8** → **3**

5 inputs — ReLU hidden layers — Sigmoid

$$(p_0, p_1, p_2) = \mathrm{NN}_\theta(x')$$

Training: continuous keep weights

$$s_k = 1 - p_k$$

Inference: keep or reject a candidate

High probability → reject the candidate.

> The network changes candidate decisions; the solver still determines fluxes and time evolution.

Physics training evaluates the evolved solution

01 Imitation learning

Learn the classical TENO5 keep/discard labels.

$$p = \mathrm{NN}_\theta(x')$$

Target: classical TENO5 decisions.

Useful initialization; the teacher supplies no direct signal for lower physical solution error.

02 Unrolled physics training

Differentiate through K complete SSPRK3 steps.

$$U_{\mathrm{pred}} = \mathcal{S}_\theta^K(U_{\mathrm{start}})$$

Compare the predicted solution with the physical reference.

The gradient crosses reconstruction, fluxes and every Runge–Kutta substage.

Training objective, in simple terms

$$\mathcal{L} = \text{prediction error} + \text{stability penalties}$$

> Pretrain by imitation, normalize loss terms, then fine-tune against a physical reference.

Stable training follows a staged procedure

Start from a reliable reconstruction, then increase the physical training horizon.

01 Pretrain

Learn TENO5 decisions before physical training.

02 Curriculum

Extend the rollout from the previous stage.

03 Guard states

Check physical states at every SSPRK3 stage.

04 Validate

Select the best model on the full benchmarks.

Euler rollout curriculum

$$K: \quad 2 \longrightarrow 4 \longrightarrow 8 \longrightarrow 15$$

Each stage starts from the previous stage's weights.

Bound the optimizer's gradient

Clip very large gradients before updating the network with Adam.

This prevents a single large update from disrupting training.

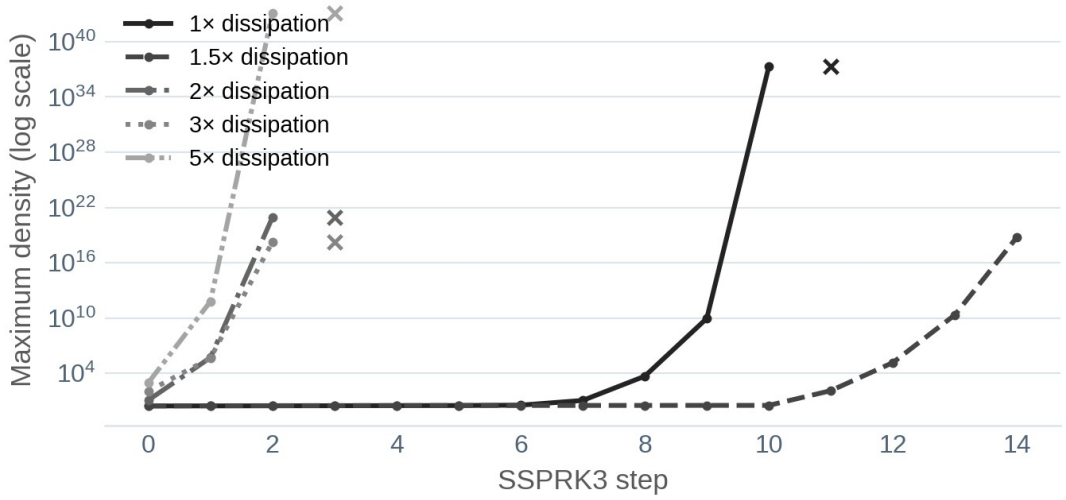
> Short-horizon training must be checked again at the full benchmark horizon.

Hybrid reconstruction supports stable Euler training

$$\Phi_{\mathrm{mix}} = (1-\lambda)\Phi_{\mathrm{TENO5}} + \lambda\Phi_{\mathrm{NN}}, \qquad 0 \le \lambda \le 1$$

The mixing head learns how much to trust the neural correction.

Observed Euler training instability



× marks a nonfinite state. Extra dissipation alone did not stabilize training.

Physical admissibility

$$\rho > 0, \qquad p > 0$$

Density and pressure must remain positive.

Guard every substage

Measure positivity violations before repair.

Control optimization

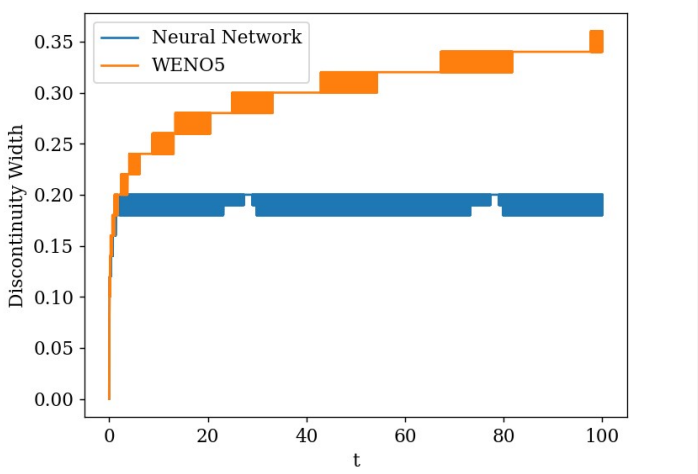
Use rollout curriculum, anchoring and clipping.

Validate the final solver

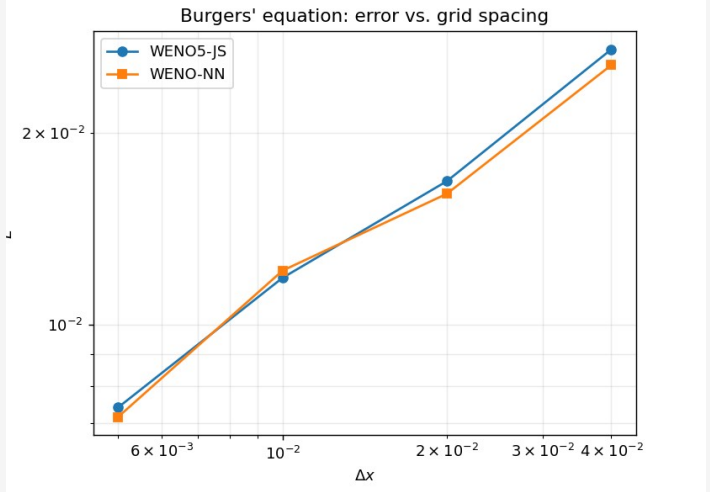
Mixing alone is not a positivity or TVD proof.

Final scalar benchmarks improve on TENO5

Earlier WENO-NN experiments



Sharper advected discontinuities



Competitive Burgers convergence

L_2 measures solution error against a reference. Lower is better.

Final gains come from the dedicated learned mixing head (Phase 19).

Final NN-TENO5

Dedicated mixing head · Phase 19

3.5%

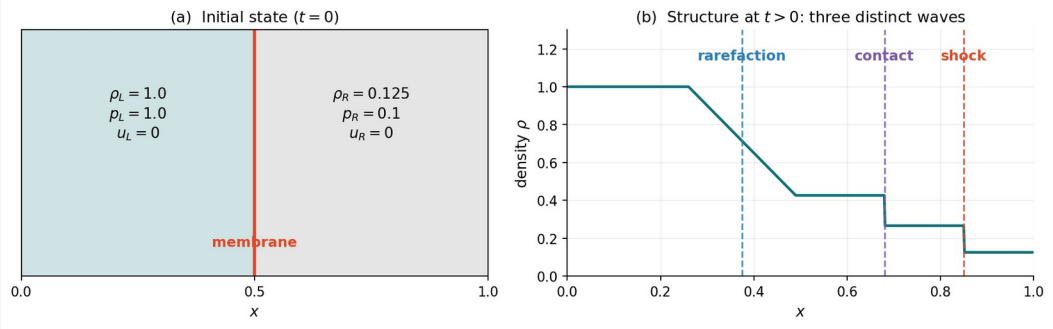
lower L_2 error on advection

18.6%

lower L_2 error on Burgers

Euler gains depend on the benchmark

Sod shock tube

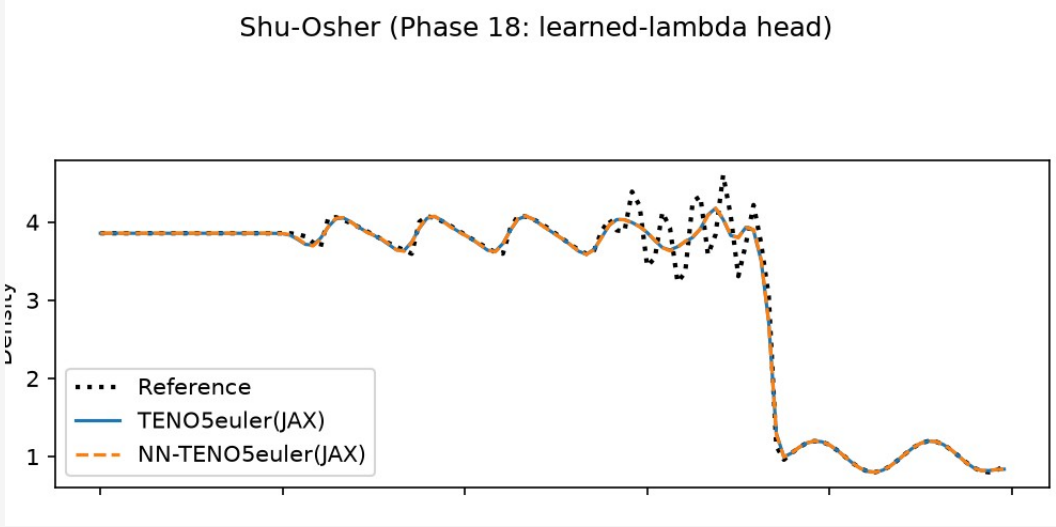


Rarefaction, contact discontinuity and shock

+0.35% error

Near parity; TENO5 remains slightly better.

Shu–Osher shock–entropy interaction



−1.84% error

A modest gain with the learned mixing head.

Metric: mean L_1 over (ρ, ρv, E). 120 cells, CFL 0.4. Sod: exact reference; Shu–Osher: 320-cell TENO5 reference.

Canonical benchmarks quantify the final gains

Learned convex mixing compared with classical TENO5 on the same benchmark configuration.

BENCHMARK	ERROR METRIC	TENO5	NN-TENO5	RATIO R	CHANGE IN ERROR
Advection	Scalar L_2	9.8960×10^{-2}	9.5521×10^{-2}	0.9652	**-3.48%**
Burgers	Scalar L_2	8.1670×10^{-4}	6.6493×10^{-4}	0.8142	**-18.58%**
Sod	Mean Euler L_1	8.2583×10^{-3}	8.2871×10^{-3}	1.0035	**+0.35%**
Shu–Osher	Mean Euler L_1	2.1319×10^{-1}	2.0926×10^{-1}	0.9816	**-1.84%**

$$R = \frac{\mathrm{error}_{\mathrm{NN}}}{\mathrm{error}_{\mathrm{TENO5}}}, \qquad \mathrm{gain}(\%) = 100(1 - R)$$

Green: lower error. Red: higher error. Euler uses mean L_1 over (ρ, ρv, E); scalar results use L_2.

Learning adds value inside a robust numerical solver

Learn the reconstruction decision.
Keep the numerical and physical structure visible.

What I built

WENO/TENO reconstruction framework

Neural stencil-decision models

Differentiable CFD training in JAX

What the results show

Scalar gains: 3.5% and 18.6% in L_2

Shu–Osher: 1.84% lower mean L_1

Sod: near parity, 0.35% higher error

What comes next

Improve the Euler reconstruction model

Validate across grids and flow cases

Extend the framework toward 2D CFD

Discussion

Thank you

Questions & discussion